THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · NCERT-TYPE CHAPTER TEST SET 2

Chapter 8: **Playing with Constructions** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The diagonals of a rectangle are each 16 cm long and meet at O. The length OA (cm) is: 1

(a) 7 **(b)** 8 **(c)** 9 **(d)** 16

2. In rectangle ABCD, diagonals meet at O and $\angle AOB = 80^\circ$. Then $\angle OAB$ is (degrees): 1

(a) 40 **(b)** 50 **(c)** 100 **(d)** 60

3. A circle of radius 5 cm and a concentric circle of radius 9 cm are drawn. The width of the ring between them (cm): 1

(a) 3 **(b)** 5 **(c)** 6 **(d)** 4

4. The diagonals of a rectangle that is not a square: 1

(a) are unequal **(b)** are equal but not perpendicular
(c) are perpendicular **(d)** are parallel

5. A square of side 6 cm: the length of the segment joining mid-points of two opposite sides (cm) is: 1

(a) 7 **(b)** 9 **(c)** 5 **(d)** 6

6. The perimeter of a rectangle with sides 13 cm and 9 cm is (cm): 1

(a) 53 **(b)** 31 **(c)** 44 **(d)** 57

7. A circle has centre O and radius 6 cm. A point P is 4 cm from O. The point P lies: 1

(a) outside the circle **(b)** on the circle
(c) inside the circle **(d)** at the centre of the circle

8. Two points A and B are 18 cm apart. The perpendicular bisector of AB meets AB at a point which is at what distance from A (cm)? 1

(a) 8 **(b)** 9 **(c)** 10 **(d)** 18

9. The diagonals of a rectangle are each 20 cm long and meet at O. The length OA (cm) is: 1

(a) 20 **(b)** 11 **(c)** 10 **(d)** 9

10. In rectangle ABCD, diagonals meet at O and $\angle AOB = 100^\circ$. Then $\angle OAB$ is (degrees): 1

(a) 80 **(b)** 40 **(c)** 50 **(d)** 90

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. In rectangle ABCD the diagonals meet at O. If AC = 14 cm, find OA, OB, BD. 2

12. AB = 10 cm. Describe how to find the point equidistant from A and B on AB and its distance from A. 2

13. A circle has radius 5 cm. A point P is 9 cm from its centre. Is P inside, on or outside the circle? Give the reason. 2

14. In rectangle ABCD the diagonals meet at O. If AC = 18 cm, find OA, OB, BD. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Describe the construction of the perpendicular bisector of a line segment of length 8 cm. 3

16. In a square with diagonal intersection O and side 6 cm, state the angle between the diagonals and the angle $\angle OAB$. 3

17. A circle has radius 8 cm. Find the lengths of its diameter and of its longest chord. Can there be a chord 17 cm long? 3

OR

Describe the construction of the perpendicular bisector of a line segment of length 9 cm. 3

18. In a square with diagonal intersection O and side 7 cm, state the angle between the diagonals and the angle $\angle OAB$. 3

OR

A circle has radius 10 cm. Find the lengths of its diameter and of its longest chord. Can there be a chord 21 cm long? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Construct a rectangle with sides 6 cm and 4 cm using only a ruler and compass (describe the steps). Verify by measuring that the diagonals are equal (state the property used). 5

OR

A circle has centre O and radius 5 cm. Points P, Q, R are at distances 5 cm, 3 cm and 7 cm from O. State whether each lies inside, on or outside the circle, and how many lie inside or on the circle. 5

20. Construct a rectangle with sides 15 cm and 8 cm using only a ruler and compass (describe the steps). Verify by measuring that the diagonals are equal (state the property used). 5

OR

A circle has centre O and radius 7 cm. Points P, Q, R are at distances 9 cm, 7 cm and 2 cm from O. State whether each lies inside, on or outside the circle, and how many lie inside or on the circle. 5

END OF QUESTION PAPER — Total: 40 Marks